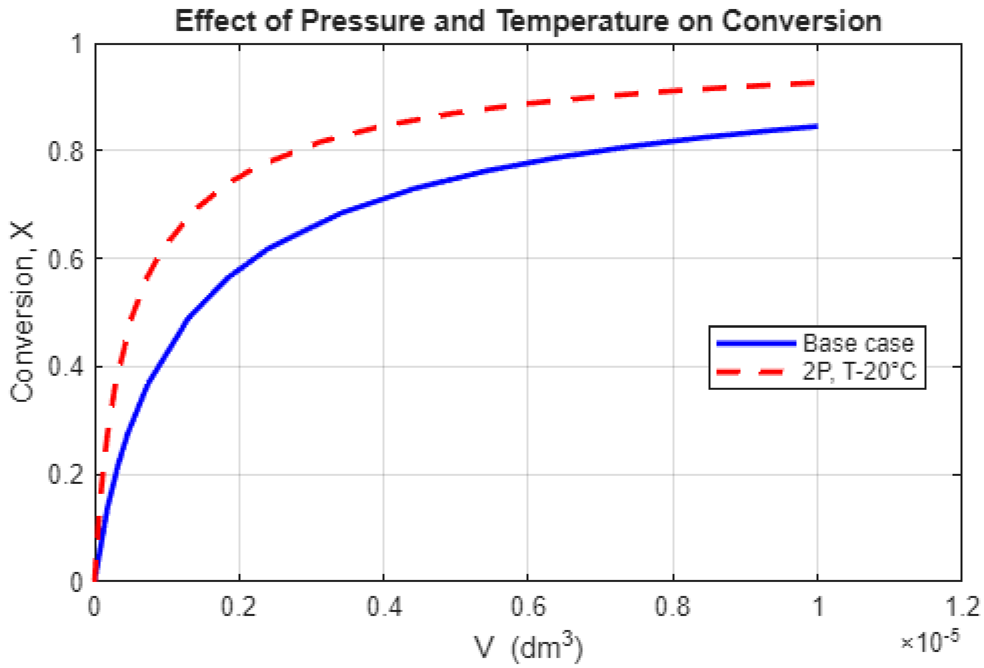
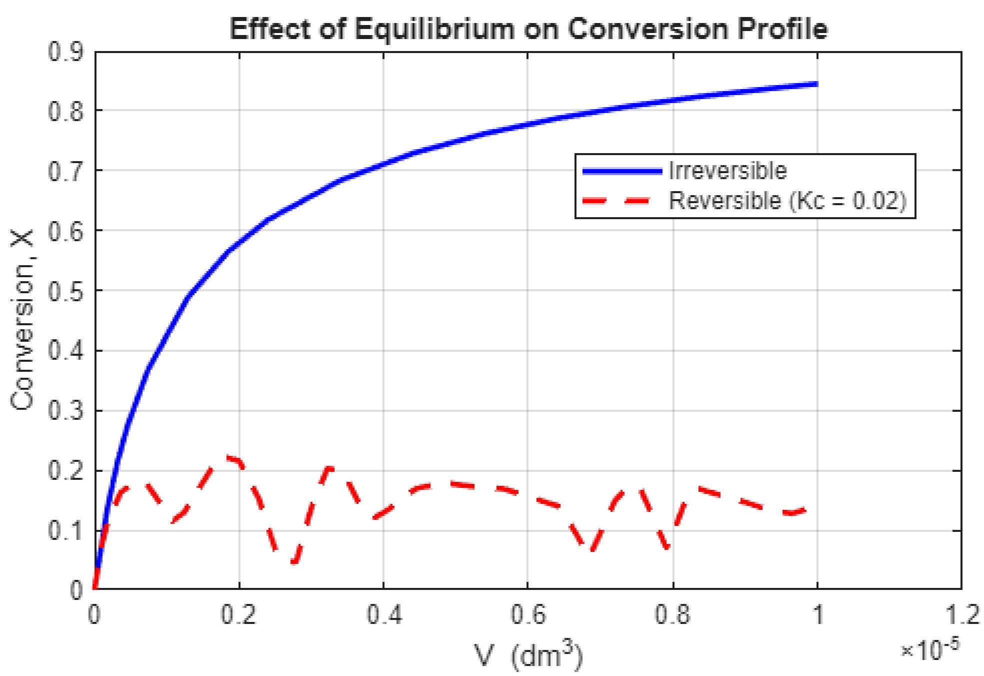
P6-1B

a) (1) The pressure doubling increases concentration, increasing rate of reaction, while lowering temperature decreases k, decreasing rate of reaction. Ending up with a conversion rate displayed in the graph produced by matlab.

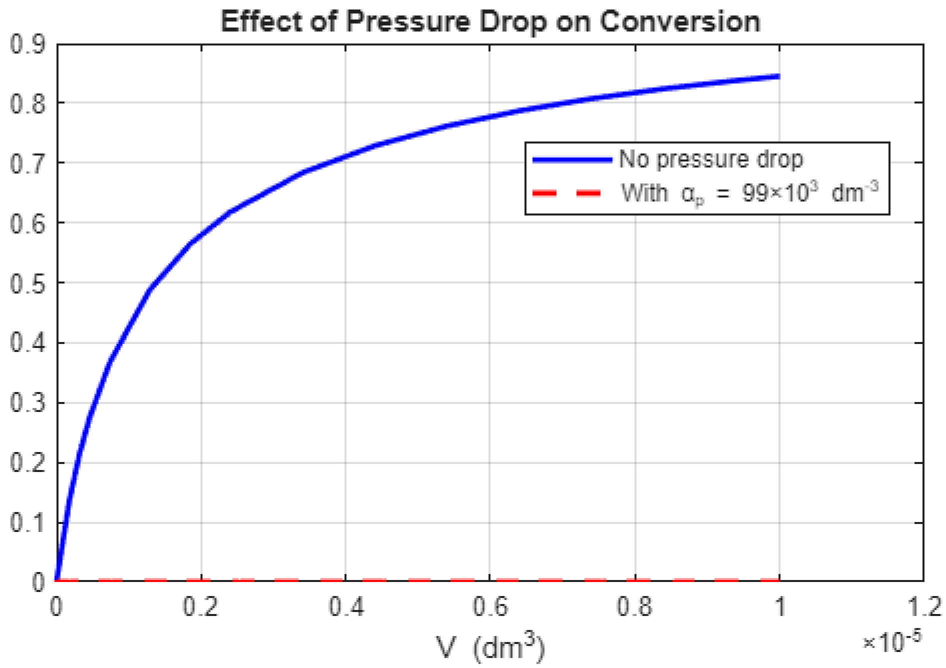


(2) The matlab file produced this graph showing the vast differences between the reversible and irreversible cases.The conversion graph for the irreversible case follows a logarithmic trend and for the reversible trend doesn't follow any particular trends.(3)(

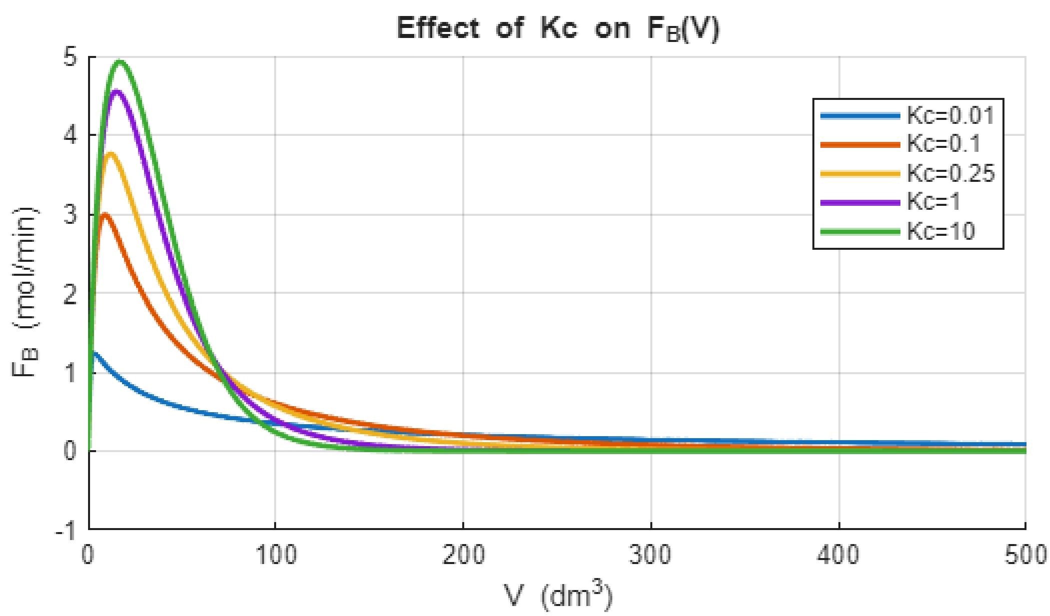


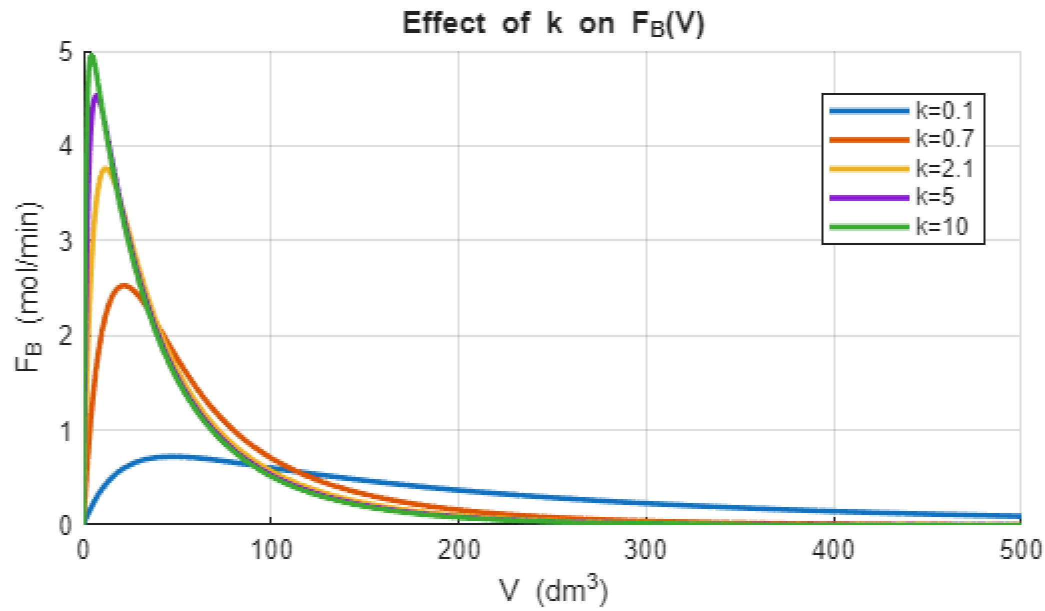
(3) The matlab file produced this graph comparing the irreversible case with no pressure

drop and with a pressure drop present. It shows that conversion rate plummets remaining zero as volume increases.

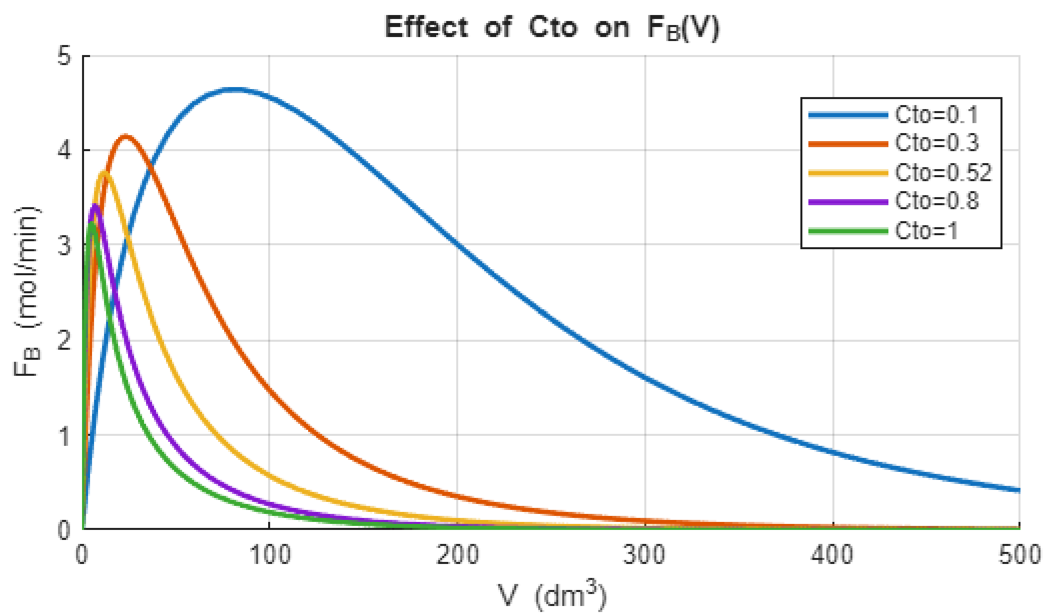


b) (1)

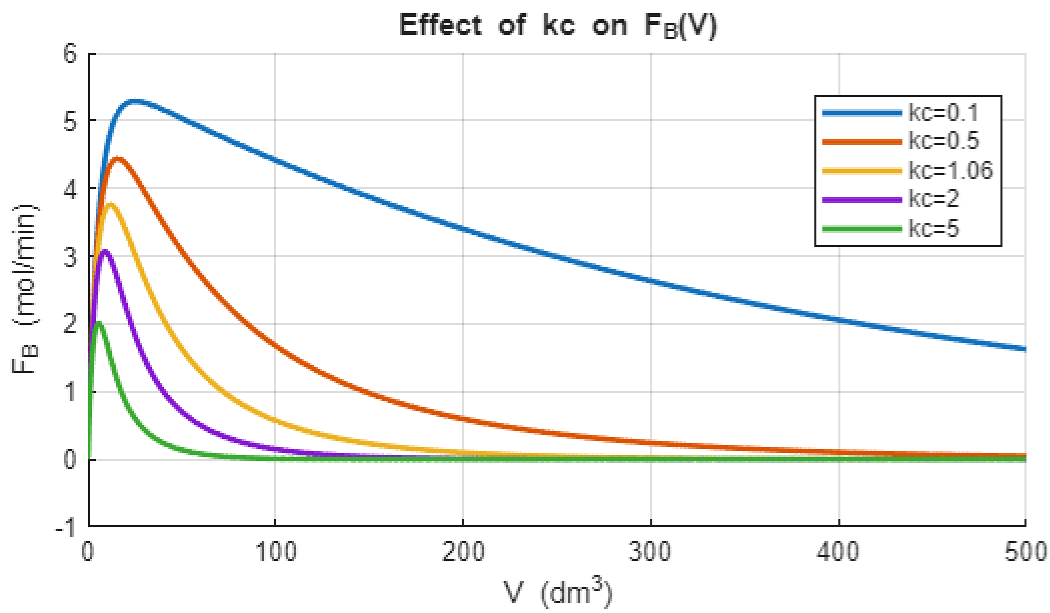




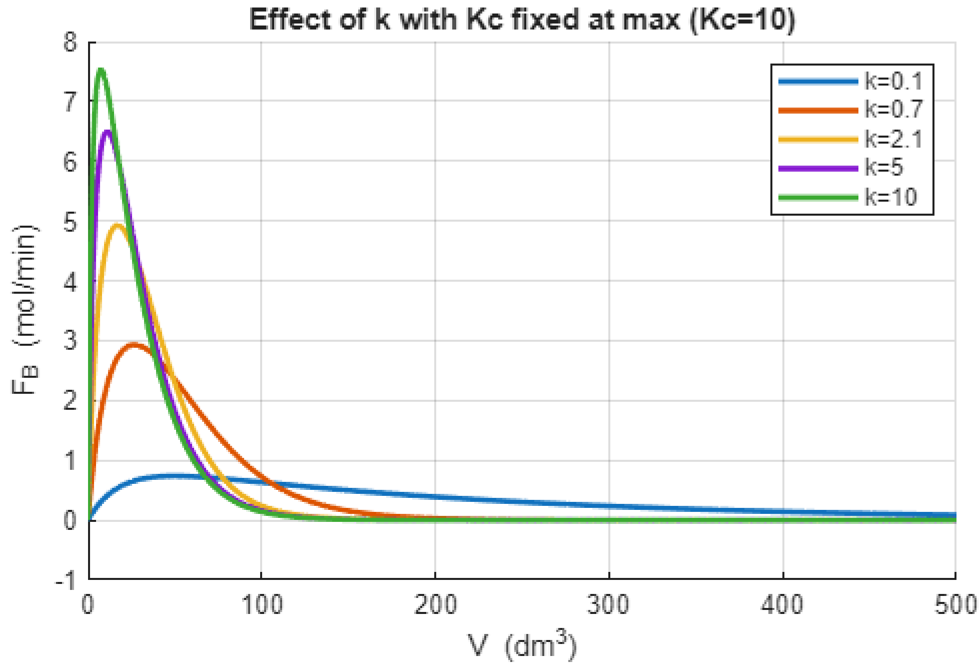
Effect of k on F_B(V)
k=0.1
k=0.7
k=2.1
k=5
k=10
F_B (mol/min)
V (dm^3)

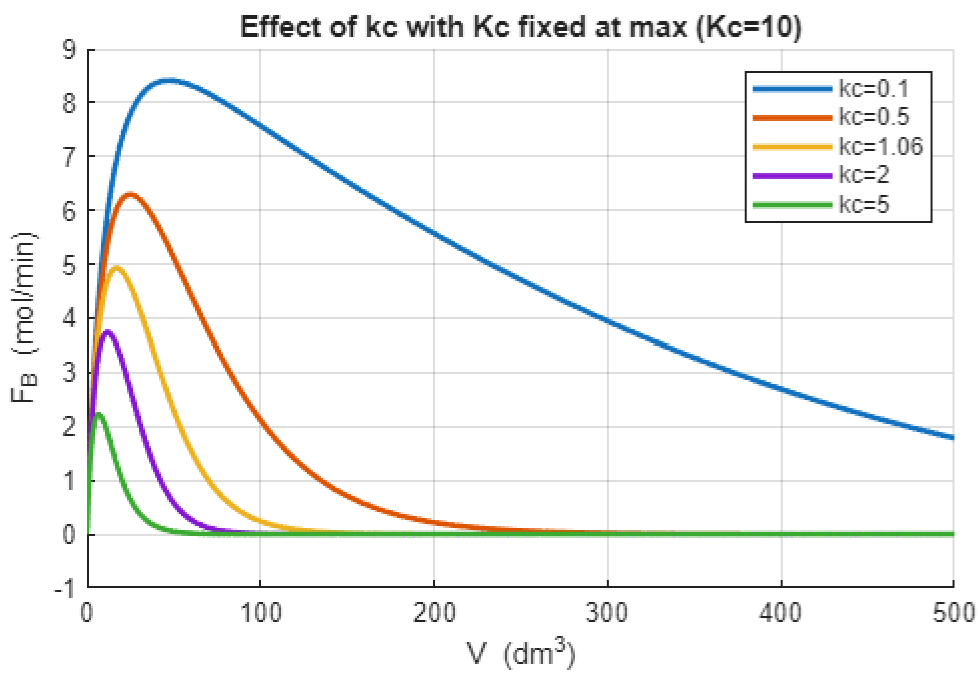


Effect of Cto on F_B(V)
Cto=0.1
Cto=0.3
Cto=0.52
Cto=0.8
Cto=1
F_B (mol/min)
V (dm^3)


Effect of kc on F_B(V)
kc=0.1
kc=0.5
kc=1.06
kc=2
kc=5
F_B (mol/min)
V (dm^3)
6
5
4
3
2
1
0
-1
0
100
200
300
400
500

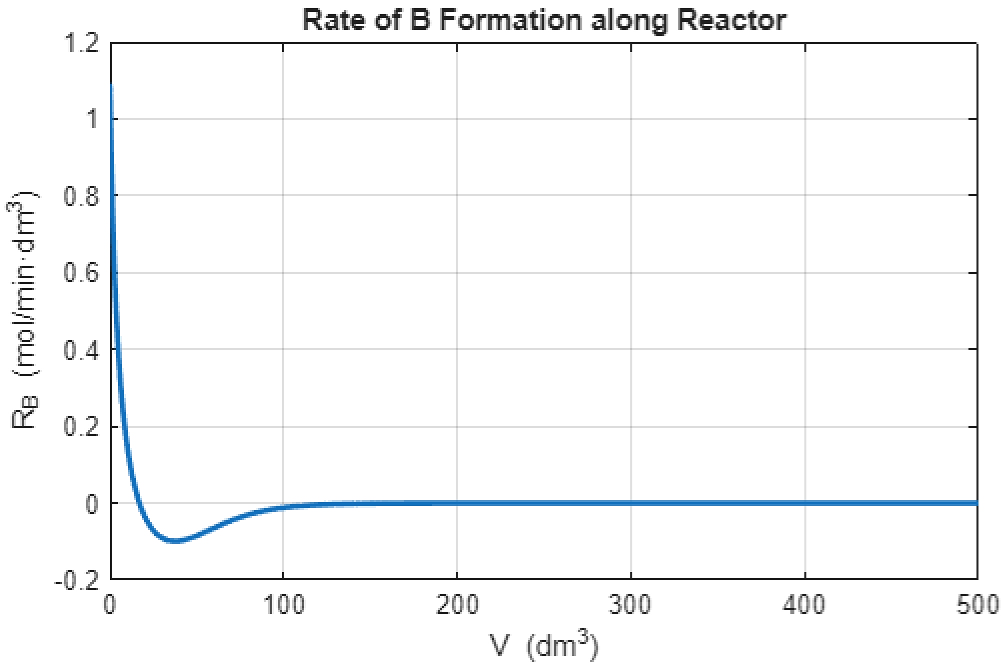
(2)


Effect of k with Kc fixed at max (Kc=10)
k=0.1
k=0.7
k=2.1
k=5
k=10
F_B (mol/min)
V (dm^3)
8
7
6
5
4
3
2
1
0
-1
0
100
200
300
400
500

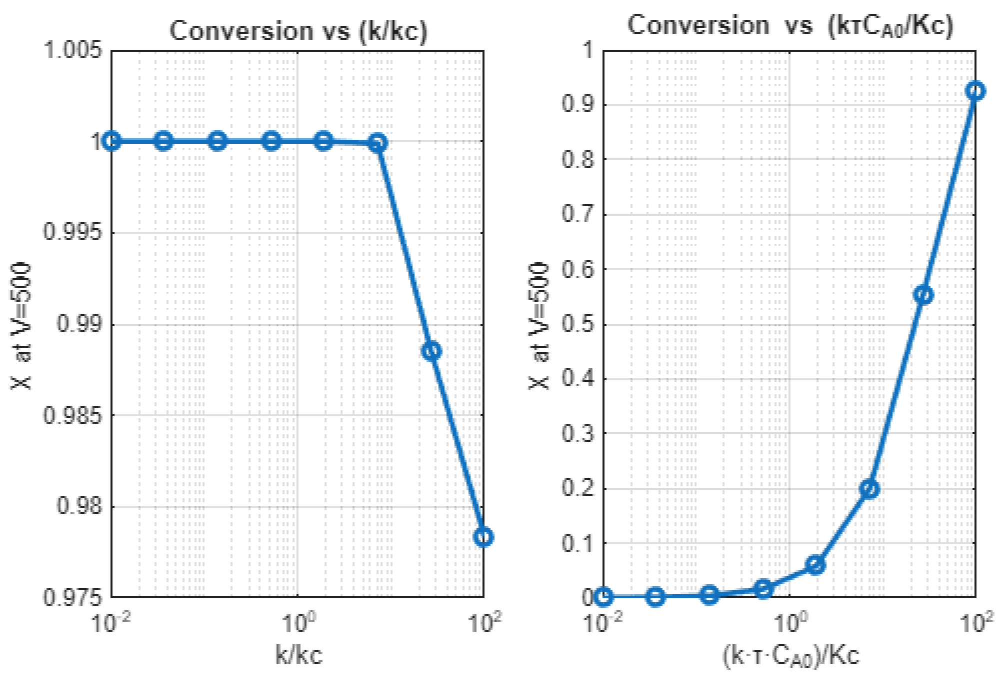

Effect of kc with Kc fixed at max (Kc=10)
kc=0.1
kc=0.5
kc=1.06
kc=2
kc=5
F_B (mol/min)
V (dm^3)
9
8
7
6
5
4
3
2
1
0
-1
0
100
200
300
400
500

(3)

```
Maximum RB = 1.0920 at V = 0.0 dm^3
```



(4)

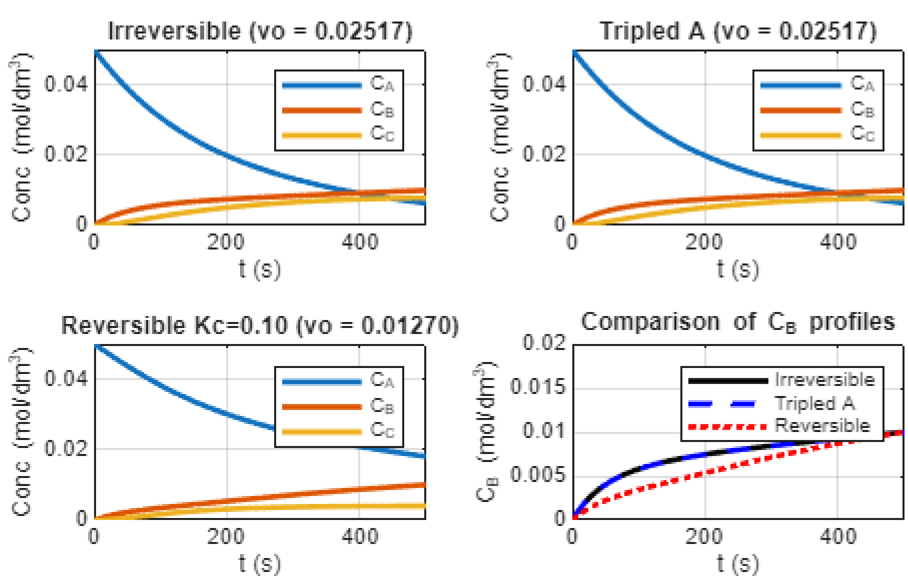


The graph showing the relationship between conversion and k/kc has a constant conversion before following a linear trend downwards while the graph showing the relationship between conversion and ktca0/kc has a exponential increase in conversion. So, the second ratio of parameters has a greater impact on X.

(5) Several clear trends were observed. Increasing the reaction rate constant k generally increased the formation rate of product B and the overall conversion X, with diminishing returns at very high k where the reaction became limited by the equilibrium constant KC and membrane removal rate kC. Increasing kC, the membrane permeation constant, initially enhanced conversion by removing B from the reactor and driving the reaction forward, but excessively high kC decreased the residence time of reactants, slightly reducing the final X. Higher values of the equilibrium constant KC shifted the reaction toward products, increasing both FB and X, while low KC limited product formation, even when k was high. Varying the total feed concentration CT0 showed that higher reactant availability increased the overall rate and conversion, though the effect was smaller than that of k or KC..Analysis of parameter ratios revealed that the ratio k/kC had the most significant effect on final conversion: when reaction and membrane removal were balanced (moderate k/kC), conversion peaked, while very high or very low ratios caused suboptimal performance. The ratio (kτCA0/KC)also affected conversion but less dramatically, indicating that both reaction kinetics and equilibrium drive product formation, but the balance between reaction and membrane flux dominates reactor performance. Overall, the study highlights that reactor design must carefully balance reaction rate, membrane permeability, and equilibrium considerations to maximize both product formation and conversion.

(6)

c) (1) The maximum feed rate of B would be Cao = 0.050 -> max vo = 0.025165 dm^3/s.
(2) The maximum feed rate of B would not change, remaining constant as A is tripled Cao = 0.150 -> max vo = 0.025165 dm^3/s
(3) The maximum feed rate of B would be Cao = 0.050, Kc = 0.10 -> max vo = 0.012696 dm^3/s when reversible.



d) After running the simulation for atrazine degradation along the reactor length, I varied several parameters—rainfall (represented by Q), evaporation rate (which indirectly affects v through water balance), inlet atrazine concentration (CA0), and liquid flow rate (v0). Increasing rainfall (Q) adds more water to the system, which increases the volumetric flow rate v and dilutes the atrazine concentration, leading to a lower overall conversion along the reactor length. Conversely, a higher evaporation rate reduces v, concentrating atrazine and increasing conversion because the reaction rate depends on C_A. When the inlet atrazine concentration (C_{A0}) was raised, the model predicted a faster initial rate of degradation due to the higher driving concentration, but the fractional conversion X decreased slightly since more total atrazine entered the system. Increasing the initial liquid flow rate (v_0) shortened the residence time, which reduced conversion because the fluid spent less time in contact with the catalyst bed. Overall, the results show that hydrodynamic and concentration changes significantly affect the extent of atrazine removal, emphasizing the importance of controlling water input and flow conditions in environmental reactor design.

e) *Cooling rate:* faster cooling produces a larger, shorter nucleation burst (higher peak r_N), yields many small nuclei, and, if residence time is long, increases coagulation later; slower cooling yields fewer nuclei but larger particles by condensation. *Flow rate:* higher flow (shorter residence time) reduces both condensation and coagulation so you see higher N retained but smaller final d_p; lower flow increases condensation and coagulation giving larger d_p and fewer particles. *N(t) shape:* both gases show a rapid rise (nucleation) then leveling/decline (coagulation). With He the nucleation peak tends to occur earlier and the late-time N is higher (weaker coagulation), so the He curve rises earlier and decays less than Ar. *Final dp:* typically smaller with He than with Ar because He's much lower gas density reduces the coagulation sink (less merging), leaving more particles that share condensed volume. Condensation is faster with He (increasing d_p) but usually not enough to overcome the coagulation reduction — net: $d_p(He) < d_p(Ar)$ in most cases. *Nucleation peak timing:* occurs earlier with He due to faster monomer removal (higher diffusion / kinetic correction) which shortens the nucleation window.

f) Question: A chemical engineer is comparing a gas-phase reaction carried out in a PFR with the same reaction carried out in a series of CSTRs. The engineer observes that the CSTRs require more total volume than the PFR to achieve the same conversion. Instead of accepting this result, identify and explain at least three assumptions in the models for PFR and CSTRs that could affect this conclusion. How might relaxing each assumption change the design choice? It uses critical thinking because you have to analyze and compare reactor models and you must use reasoning and use knowledge of several models.

P6-5B

a) $k=k_0\exp(-\frac{E_a}{R}(1/T-1/T_0))$
$1/T-1/T_0=1/300-1/323$
$k = 8.9\times10^{-6}\ min^{-1}$
$r_A=k(C_A-C_B C_C^2/K_c)$
$C_A-C_B C_C^2/K_c = 0$
$C_B=C_{A0}\cdot X, C_C=2C_{A0}\cdot X$
$X_{eq}=0.52$
$dF_A/dV=r_A$
$F_{A0}dX/dV=r_A$
$X=0.47$

b) .

```matlab
% membrane_reactor.m
% Simulate A -> B + 2C in a plug flow with membrane removal of C
clear; close all; clc;
% ---------- Input data ----------
P_atm    = 10;           % atm
R_latm   = 0.082057;     % L·atm/(mol·K)
T_reactor= 300;          % K for membrane case (given)
FA0      = 2.5;          % mol/min (feed A)
FB0      = 0;
FC0      = 0;
% Kinetics / equilibrium
k_ref    = 1e-4;         % min^-1 at T_ref (50 C = 323 K)
T_ref    = 323;          % K
Ea       = 85000;        % J/mol
Rj       = 8.314;        % J/mol/K
KC       = 0.025;        % dm^6/mol^2 (L^2/mol^2)
% Membrane transport
kC_s     = 0.08;         % s^-1 (given)
kC       = kC_s * 60;    % convert to min^-1
% ---------- Derived quantities ----------
CT = P_atm / (R_latm * T_reactor);   % mol/L (total concentration at P,T)
% kf at reactor T (Arrhenius)
kf = k_ref * exp( Ea/Rj * (1/T_ref - 1/T_reactor) );   % min^-1
kr = kf / KC;  % units consistent: kr * C_B * C_C^2 has units mol/L/min
% ---------- ODE system ----------
% State vector: y = [FA; FB; FC]
odefun = @(V,y) reactor_odes(V,y,FA0,kf,kr,KC,CT,kC);
% stop when conversion X >= 0.8
options = odeset('Events', @(V,y) stop_at_X(V,y,FA0,0.80), 'RelTol',1e-8,'AbsTol',1e-10);
% integrate until conversion reaches 0.8 or up to a large V
Vmax_guess = 1e5; % large upper limit in L
[Vsol, Ysol] = ode45(odefun, [0 Vmax_guess], [FA0; FB0; FC0], options);
FA = Ysol(:,1);
FB = Ysol(:,2);
FC = Ysol(:,3);
Ftot = FA + FB + FC;
X    = 1 - FA./FA0;
V    = Vsol;
% ---------- Find maxima in flows ----------
[FC_max_val, idxC] = max(FC);
V_FC_max = V(idxC);
[FB_max_val, idxB] = max(FB);
V_FB_max = V(idxB);
```

```
[FA_min_val, idxA] = min(FA); % FA typically monotonically decreases
% ---------- Plots ----------
figure(1);
plot(V, X, 'LineWidth',1.5);
xlabel('Reactor volume V (L)');
ylabel('Conversion X');
grid on;
title('Conversion vs Volume (membrane PFR)');
xlim([0 V(end)]);
figure(2);
plot(V, FA, '-','LineWidth',1.3); hold on;
plot(V, FB, '--','LineWidth',1.3);
plot(V, FC, ':','LineWidth',1.3);
xlabel('Reactor volume V (L)');
ylabel('Molar flow (mol/min)');
legend('F_A','F_B','F_C','Location','best');
title('Molar flows vs Volume (membrane PFR)');
grid on;
% annotate maxima
plot(V_FC_max, FC_max_val, 'ko','MarkerFaceColor','k');
text(V_FC_max, FC_max_val, sprintf('  F_C_{max}=%.3g at V=%.1f L', FC_max_val,
V_FC_max));
plot(V_FB_max, FB_max_val, 'ks','MarkerFaceColor','k');
text(V_FB_max, FB_max_val, sprintf('  F_B_{max}=%.3g at V=%.1f L', FB_max_val,
V_FB_max));
% ---------- print summary ----------
fprintf('kf (min^-1) = %.4e  ; kr (min^-1 * (mol/L)^{-2}) = %.4e\n', kf, kr);
fprintf('CT (mol/L) = %.4f\n', CT);
fprintf('Membrane kC (min^-1) = %.3f\n', kC);
fprintf('Stopped at V = %.3g L, conversion X = %.4f\n', V(end), X(end));
fprintf('F_C max = %.4f mol/min at V = %.3f L\n', FC_max_val, V_FC_max);
fprintf('F_B max = %.4f mol/min at V = %.3f L\n', FB_max_val, V_FB_max);
% ---------- nested functions ----------
function dydV = reactor_odes(~, y, FA0, kf, kr, KC, CT, kC)
  FA = y(1); FB = y(2); FC = y(3);
  Ftot = FA + FB + FC;
  % avoid divide-by-zero
  if Ftot <= 0
     CA = 0; CB = 0; CC = 0;
  else
     CA = (FA./Ftot) * CT;
     CB = (FB./Ftot) * CT;
     CC = (FC./Ftot) * CT;
  end
```

```
  % net forward rate (mol/L/min)
  rnet = kf*CA - kr*CB*(CC.^2);
  % stoichiometric derivatives (nu_A = -1, nu_B = +1, nu_C = +2)
  dFA_dV = -rnet;
  dFB_dV = +rnet;
  dFC_dV = +2*rnet - kC*CC;   % membrane removal modeled as first-order in C concentration
  dydV = [dFA_dV; dFB_dV; dFC_dV];
end
function [value, isterminal, direction] = stop_at_X(~, y, FA0, X_target)
  FA = y(1);
  X = 1 - FA/FA0;
  value = X - X_target;      % event when value==0
  isterminal = 1;            % stop the integration
  direction = +1;            % detect only increasing through target
end
```

c)
What if the membrane transport coefficient kC were doubled?

Doubling kC would remove ethane faster from the reactor, which would in turn increase the driving force for the forward reaction, so the conversion of A could increase faster. This may also cause flow rates of C to peak at a higher value earlier.